

Negative Exponent LawQuestion Bank 3.04 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 6, rewrite each expression as an equivalent expression with positive exponents. Do not evaluate.**

1. 8^{-3}

$\frac{1}{8^3}$

2. $\frac{1}{(-4)^{-5}}$

$(-4)^5$

The power moves up and the exponent turns positive; the base keeps its sign.

3. $(-2)^{-8}$

$\frac{1}{(-2)^8}$

Keep the parentheses.

4. $\frac{1}{10^{-2}}$

10^2

5. $-5 \cdot 6^{-4}$

$-5 \cdot \frac{1}{6^4} = -\frac{5}{6^4}$

The -5 is a factor, not a base with an exponent; it stays where it is.

6. $\frac{2^{-3}}{9^{-2}}$

$\frac{9^2}{2^3}$

Each power swaps sides.

For questions 7 and 8, apply the negative exponent law, then the product or quotient of powers law, and evaluate.

7. Evaluate each expression.

a. $(-3)^2 \cdot (-3)^{-5}$
 $(-3)^{-3} = \frac{1}{(-3)^3} = -\frac{1}{27}$

b. $2^8 \cdot 2^{-4}$
 $2^4 = 16$

c. $9^9 \cdot 9^{-9}$
 $9^0 = 1$

8. Evaluate each expression.

a. $(-5)^{-6} \div (-5)^{-4}$
 $(-5)^{-2} = \frac{1}{25}$
 $-6 - (-4) = -2$; an even power of a negative is positive.

b. $2^0 \div 2^{-3}$
 $2^3 = 8$

c. $3^{-7} \div 3^5$
 $3^{-12} = \frac{1}{3^{12}} = \frac{1}{531,441}$

Opposite or reciprocal.

9. Four expressions are shown: -4 , 4^{-1} , $(-4)^{-1}$, -4^{-1} .

a. Give the value of each one.

-4 ; $1/4$; $-1/4$; $-1/4$

4^{-1} is the reciprocal of 4. $(-4)^{-1}$ is the reciprocal of -4 . -4^{-1} is the opposite of the reciprocal of 4 — the same value as $(-4)^{-1}$ here, for a different reason.

b. Which of the four is the reciprocal of 4, and which is the opposite of 4?

4^{-1} is the reciprocal of 4; -4 is the opposite of 4.

Working backwards.

10. Find the value of n that makes each statement true.

a. $2^n = \frac{1}{32}$

$n = -5$

b. $10^n = 0.001$

$n = -3$

c. $7^3 \cdot 7^n = 7^{-1}$

$n = -4$

$3 + n = -1.$

11. Which is the value of 4^{-3} ?

☒ A. $\frac{1}{64}$

☐ B. -64

☐ C. -12

☐ D. $\frac{1}{12}$

A

12. Select ALL of the expressions whose value is $\frac{1}{64}$.

☒ A. 2^{-6}

☒ B. 4^{-3}

☒ C. 8^{-2}

☒ D. $\left(\frac{1}{2}\right)^6$

☐ E. -2^6

☐ F. 2^{-8}

A, B, C and D